

Jacob Stephens

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➦ [Portfolio](#) |  [GitHub](#) | in [LinkedIn](#)

Senior full-stack engineer and lead technology owner for a multi-million-dollar specialty travel operator — architecting and operating the reservations platform, payment systems, Linux server fleet, and developer tooling the business runs on. Stack spans PHP, TypeScript, and Python on MySQL, with deep experience in REST APIs, PWAs, PCI-sensitive workflows, and AI-assisted developer tooling. Strong track record improving performance, reducing operational risk, and shipping business-critical product features.

Core Technologies

Languages & Databases

- [PHP](#)
- [MySQL](#)
- [JavaScript](#)
- [HTML](#)
- [CSS](#)

Shipped with AI-Assisted Tooling

- [Rust](#)
- [Node.js / Express](#)
- [TypeScript](#)
- [Python](#)
- [Flask](#)
- [FastAPI](#)
- [PostgreSQL](#)
- [SQLite](#)

Frameworks & Platforms

- [Progressive Web Apps](#)
- [React](#)
- [WordPress](#)

Engineering Practices

- [AI-Assisted Development \(Claude Code, MCP\)](#)
- [PCI Compliance](#)

APIs & Application Engineering

- [REST APIs](#)
- [API Integration \(Twilio, Textable\)](#)
- [Queue Systems & Automation](#)
- [ACH/NACHA File Workflows \(SFTP\)](#)
- [Automated Testing \(Playwright\)](#)
- [Encryption & Privacy-First Design](#)
- [Performance Optimization](#)
- [HMAC-signed webhooks](#)

Infrastructure & Operations

- [Linux Administration](#)
- [Apache](#)
- [Caching Infrastructure](#)
- [Offline Sync](#)
- [Git](#)
- [DNS Management](#)
- [Caddy](#)
- [Docker / Docker Compose](#)

Work

Senior Infrastructure Developer and Full Stack Engineer

[Educational Travel Adventures](#) | Feb 2022 - Present | United States, Remote

- Serve as lead engineer and continuity owner for the company's production stack — reservations platform, payment processing, Linux server fleet, and deployment pipeline — with end-to-end responsibility for architecture, operations, and on-call response across a multi-million-dollar travel business
- Built guides.etadventures.com — a mobile-first PWA for field contractors with service-worker offline support, live auto-save reporting, Twilio-powered SMS to travelers, receipt-attached expense workflows, and manager-side administration — extending the JavaScript stack across a previously PHP-only platform
- Architected automated reservation cancellation workflows in PHP and MySQL, preventing bad debt accumulation with business-rule-driven automation that protects revenue and improves cash flow management
- Built payment notification queue management system with Twilio SMS, transactional email, and voice capabilities, enabling self-service configuration and reducing manual work by 80%+ while ensuring timely customer communication
- Optimized alert system performance through caching infrastructure and database query improvements, reducing page load times from 5-10 seconds to under 1 second and reducing database load by 70-80%
- Optimized Tourbot's group manifest page through profile-driven N+1 elimination, batched in-request prefetch caches, and a targeted history-table schema and index change, reducing page load from 5-7 seconds to ~1 second

- and SQL statements per request from ~2,650 to ~183 with byte-identical HTML output
- Led PCI compliance initiative, architecting secure REST integrations with TranSafe and Monetra merchant processors and ACH/NACHA file workflows over SFTP to the bank, implementing security best practices including bank-account encryption
- Led three enterprise-wide platform migrations (PHP 5 -> 8, MySQL 5 -> 8, and CentOS 7 -> Rocky Linux 9) across the codebase and server fleet, modernizing the stack and improving application performance
- Established engineering practices across internal and customer-facing platforms — owning architecture decisions, code review, and mentorship for the development team
- Built an internal orchestration and status platform in Python, Flask, and SQLite: a Caddy-fronted dashboard at status.etadventures.com that probes the company's MySQL, web, and cron servers every 60 seconds, plus a hardened service-account pipeline (1Password-injected secrets, IP-pinned MySQL grants, sudo-scoped wrapper) that lets a Claude Code agent operate across multiple servers with auditable, scoped credentials

Web Developer and Client Support Specialist

[Sharp Innovations](#) | Nov 2021 - Feb 2022 | [Conestoga, PA](#)

- Developed automated database migration scripts using SQL and PHP, reducing migration time from hours to minutes and eliminating manual errors, improving deployment efficiency and reliability
- Delivered comprehensive technical support for web clients, troubleshooting and resolving complex issues including UI/UX updates, security vulnerabilities, and Linux server configuration and optimization
- Implemented project management task board system to replace email-based workflow, improving team productivity, project visibility, and cross-functional collaboration

Full-Stack Developer (Independent)

[Steward Goods](#) | Mar 2020 - Nov 2021 | United States, Remote

- Designed and shipped artifact.stewardgoods.com, a multi-tenant possessions tracker built on full-stack PHP and MySQL with frontend, backend, and deployment ownership
- [Optimized](#) website performance through code and infrastructure improvements, achieving perfect scores (100/100) on Google Core Web Vitals
- [Engineered](#) PCI-compliant subscription payments with Stripe in PHP and MySQL

Selected Projects

- [CreightonTracker](#): Privacy-focused fertility tracking app with encrypted sync, offline support, chart/calendar views, and provider sharing — adopted by real users who found it through organic search (creightontracker.com)
- [Cascade](#): Focus/sleep sound player built on one headless Rust core that drives native shells over a single JSON boundary. Shipped the web (PWA) and Android shells through agent-assisted development — the Rust core bound via wasm-bindgen and UniFFI — with the same core architected to extend to macOS, Windows, iOS, and watchOS (cascade.stephens.page)
- [Clowder & Crest](#): Medieval cat-guild management game in TypeScript / Matter.js with 15 distinct minigame types and Playwright end-to-end test coverage (clowderandcrest.com)
- [MacroTracker](#): Self-hosted nutrition-tracking PWA with barcode scanning, multi-API food search, recipes, CSV export, and offline-capable workflows (Node.js / TypeScript / Express on SQLite; macros.stephens.page)
- [Magisterium MCP Server](#): TypeScript MCP server exposing structured theological citations to AI assistants (magisterium.stephens.page)

Education

Bachelor of Arts in Sociology

[Gordon College](#) | Aug 2012 - Aug 2017 | [Wenham, MA](#)

LinkedIn Learning Certificates

[Full-Stack Development](#) and [WordPress Development](#) | 2020–2021 | [view](#)